

W1 Project Scope

Personalization in Federated Learning under Domain Shift

Topic 3 • Team 8-HANTA • Week 1: 04–10 May 2026

Team	Team 8-HANTA
Topic	Topic 3 — Personalization in FL under Domain Shift
Dataset	FLamby — Heart Disease (4 hospital clients, tabular)
Key Metrics	Global accuracy, average local accuracy (per client)

1. Context & Problem Statement

Federated Learning (FL) enables hospitals to train models collaboratively without sharing raw patient data. However, in practice, each hospital serves a distinct patient population and follows different acquisition protocols, causing non-IID data distributions — a problem known as domain shift. Standard FedAvg learns a single global model, which is suboptimal for individual clients under severe heterogeneity.

This project investigates whether Personalized Federated Learning (PFL) can significantly outperform both a global FedAvg model and purely local models under realistic domain shift, and what trade-offs arise in terms of convergence and communication cost.

2. Dataset

We use the FLamby Heart Disease benchmark, which partitions the UCI Heart Disease dataset across 4 real medical centers (Cleveland, Hungary, Switzerland, Long Beach), each forming one FL client. The task is binary classification (presence of heart disease) on 13 clinical features (~920 total records). Class imbalance and distributional differences between centers make this a realistic domain shift scenario. Dataset access: `pip install flamby`.

3. Objectives

- Implement FedAvg as the global FL baseline and a Local-only model as the no-collaboration lower bound.
- Implement pFedMe (primary personalized method) and optionally Per-FedAvg as a second variant.
- Compare all methods on global and average local accuracy; analyze the impact of domain shift on each approach.

4. Methodology Overview

- **Simple MLP (2–3 hidden layers) for tabular data, with separate global feature extractor and local classifier head.** Architecture:
- **Federated averaging over 4 clients — global baseline.** FedAvg:
- **Moreau envelope regularization decouples global and personalized model optimization (Dinh et al., NeurIPS 2020).** pFedMe:
- **MAML-based meta-learning for fast local adaptation (Fallah et al., 2020).** Per-FedAvg (optional):
- **80/20 per-client split, fixed seeds, 50–100 communication rounds, grid search on learning rate and regularization.** Protocol:

5. State-of-the-Art Summary

- **De-facto standard for federated learning; our global baseline.** McMahan et al. (2017) — FedAvg.
- **Personalized FL with Moreau envelopes; convergence guarantees under heterogeneity.** Dinh et al., NeurIPS 2020 — pFedMe.
- **MAML-based PFL for fast local adaptation; optional second method.** Fallah et al. (2020) — Per-FedAvg.
- **Proximal term for heterogeneous FL; considered as an additional comparison point.** Li et al. (2020) — FedProx.
- **Real-world FL healthcare benchmark providing our dataset and evaluation protocol.** Terrail et al., NeurIPS 2022 — FLamby.

6. Weekly Plan & Risk

Week	Dates	Deliverables
W1 ✓	04–10 May	Registration, Git repo, Drive folder, W1_project_scope.pdf
W2	11–17 May	Dataset access, EDA, FedAvg baseline, W2_baseline_report.pdf
W3	18–24 May	pFedMe impl., ablations, comparisons, W3_experiments_summary.pdf
W4	25–31 May	Final report (IMRAD), slides, presentation rehearsal
Final	~01 Jun	Oral presentation + optional demo

Key risks: (1) Data access delay — mitigated by pip install flamby. (2) pFedMe instability — mitigated by using reference hyperparameters from the original paper. (3) Scope creep — Per-FedAvg is optional; pFedMe is the primary focus.